3S PHARMA LOGISTICS · PACK INTELLIGENCE

Pharmaceutical Pack Size Prediction Web Application

Project Documentation

Estimate a single pharmaceutical pack's outer length, width, height, volume and weight from a trained machine-learning model — with a confidence band and the closest reference packs from your own inward history.

ExtraTrees + Gradient Boosting · Nearest-neighbour blend · 3,210+ measured packs · Confidence scored · 3D / 2D pack views

1 Executive Summary

Project Name

Pharmaceutical Pack Size Prediction Web Application

Objective

To predict the physical shipping dimensions of an individual pharmaceutical product pack — its outer length, width, height, calculated volume and approximate gross weight — directly from product attributes, using a trained machine-learning model built on the organisation's own historical inward measurement data. The tool removes the need to physically measure and weigh every new pack before it can be planned, quoted or packed.

The solution aims to

- Estimate pack dimensions and weight **before** a physical sample is in hand
- Support faster carton, courier and warehouse-space planning
- Reduce dependence on manual measuring and repeated data entry
- Turn thousands of past inward measurements into reusable operational knowledge
- Provide every estimate with a confidence score and traceable reference packs
- Offer a low-cost tool that runs locally or in the cloud

2 What is Pack Size Prediction?

Pack Size Prediction is the process of estimating the physical outer dimensions and weight of a pharmaceutical pack **automatically** from what is already known about the product — its name, brand, dosage form, strength and how many units the pack contains — instead of physically measuring each pack by hand.

Given a new product's attributes, the system automatically

1. Reads the product & pack details
2. Converts them into model features
3. Predicts L · W · H, volume & weight
4. Attaches a confidence band & range
5. Shows the closest historical packs

Example

Without prediction

A logistics executive must wait for a physical sample of a new injection vial pack, measure it with a ruler and weigh it on a scale before the carton and shipping space can be planned.

With prediction

The executive enters the product name, category (e.g. Dry Powder Injection Vial), strength and pack count, and the app instantly returns the expected length, width, height, volume and weight — plus the historical packs it learned from.

3 Why is Pack Size Prediction Necessary?

Current Challenges

Pharmaceutical logistics and packing teams routinely run into the following friction points, each of which the prediction tool is designed to remove:

Manual Measurement Effort

Every new or unrecorded pack must be physically measured and weighed before it can be planned — repetitive work that consumes staff time.

Planning Delays

Carton selection, courier booking and warehouse-space allocation stall until a physical sample is available and measured.

Inconsistent & Lost Knowledge

Thousands of measurements already exist across many workbooks, but are scattered, duplicated and rarely reused, so packs get re-measured from scratch.

Human Errors

Transposed L/W/H values, mixed units (cm vs mm), vials logged as ampoules and duplicate rows all creep in during manual entry.

Poor Scalability

As the product catalogue grows, measuring and tracking every pack manually becomes increasingly difficult and slow.

Under-used Data

Historical measurement data sits idle instead of powering faster, data-driven decisions.

4 Business Problem Statement

Existing Situation

Teams frequently need pack dimensions and weight for:

- Carton and box-size selection
- Courier and freight cost estimation
- Warehouse and shelf space planning
- Quotations and order handling
- Load and consignment planning

Today these come from physically measuring each pack — manual, repetitive and dependent on a sample being present.

Business Impact

- Delays in planning and quoting while samples are measured
- Time lost re-measuring packs measured before
- Inconsistent records and avoidable errors
- Historical measurement data left unused
- Reduced productivity on repetitive tasks

Proposed Solution

Implement a machine-learning web application that consolidates the organisation's historical inward measurements into a single curated dataset and predicts the length, width, height, volume and weight of any pack from its product attributes — instantly, with a confidence score and supporting reference packs, **before a physical sample is required.**

5 Technology Stack Evaluation & Recommendation

Three implementation approaches were evaluated. The approach that was actually **built and deployed** is presented first as the recommended option.

RECOMMENDED · IMPLEMENTED

Option 1 — Lightweight Python Web App + Local ML Model

TECHNOLOGY STACK

- Python HTTP server
- scikit-learn
- ExtraTrees + HistGB
- pandas
- openpyxl
- NumPy
- three.js
- Render

ADVANTAGES

- Low implementation & hosting cost
- Self-contained — cached model ships with the app
- Runs locally or in the cloud
- Transparent — cites the real packs it learned from
- Degrades gracefully without the ML library

User input
Product, brand, category, strength, pack counts

▼
Submit prediction form

Feature engineering
Numeric + categorical + hashed-text features

▼
Featurised query

ML model
ExtraTrees + Gradient Boosting (log-target)

▼
Blended 65 / 35 with nearest-neighbour

Prediction
L · W · H · volume · weight · confidence band

▼
Render results

Output to user
Estimate + closest reference packs + 3D / 2D drawings

Section 5 · Alternative approaches considered

Option 2 — Full Web Framework + Managed ML Service

STACK

React · FastAPI / Django · PostgreSQL · Managed ML / AutoML

ADVANTAGES

- Fully customisable UI
- Centralised multi-user data
- Role-based access & audit
- Advanced dashboards

User interface
Web app / form input

▼ *HTTP request*

Backend API
Validate, authenticate & process

▼ *Query / write*

Database
Store records, fetch training data

▼ *Send features*

ML service
Host model & return prediction

Limitations — higher build cost & longer timeline · server + DB maintenance · ongoing ML-hosting cost

Option 3 — Cloud-Native Enterprise ML Solution

STACK

Amazon SageMaker · AWS Lambda · API Gateway · Amazon S3 · CloudWatch

ADVANTAGES

- Highly scalable, enterprise-grade
- Managed retraining & versioning
- Event-driven integration
- Cost-efficient at very high volume

Request / event
API call, batch trigger, schedule

▼ *Invokes*

Lambda function
Parse request, build feature payload

▼ *Call endpoint*

SageMaker endpoint
Hosted model returns dimensions

▼ *Prediction*

Consumer
App, dashboard, ERP, WMS

Limitations — highest technical complexity · needs cloud/ML expertise · longer setup, higher fixed cost

Section 5 · Comparison & final recommendation

Comparison Table

Criteria	Python App + Local ML	Framework + ML Service	AWS Cloud-Native
Implementation Cost	Low	Medium	High
Maintenance Effort	Low	Medium	Medium
Scalability	Medium	High	Very High
Infrastructure Requirement	None / minimal	Required	Required
Customisation Capability	Medium	High	Very High
Technical Complexity	Low	Medium	High
Operational Overhead	Low	Medium	High
Transparency of Prediction	High	Medium	Medium
Business User Accessibility	High	Medium	Low
Enterprise Readiness	Medium	High	Very High

Final Recommendation — Lightweight Python Web App + Local ML Model

After evaluating all three approaches, the lightweight Python web application with a locally trained ML model is recommended because it directly addresses the business need at minimal cost, needs little to no dedicated infrastructure, runs equally well locally or on a low-cost cloud instance, keeps every prediction transparent by citing real historical packs, and can evolve into the framework or cloud-native architecture later as volumes grow.

6 How the System Works

6.1 · Data Foundation

The model is built on 3S Pharma's own historical inward records. Seven source workbooks were inspected and normalised into a single curated master dataset.

7
Source workbooks inspected

10,484
Substantive source rows preserved

7,395
Usable measured attempts

4,524
Unique measured pack profiles

3,210
Training rows · Train_Clean build

4,809
Training rows · Full_Imputed build

KEY DATA-PREPARATION RULES

- **Category replaces Form** — one clean curated field
- **Consistent units** — cm converted to mm; weight in g
- **Longest × middle × shortest** edge ordering
- **Full traceability** — source file, sheet & row kept

6.2 · DATA PIPELINE

3S_Master_ML_Ready.xlsx
Curated master workbook

▼
prepare_data.py

master_training.csv
Flat, model-ready data

▼
train_model.py

model_cache.pkl
Trained cached ensemble

▼
Loaded at startup

app.py
Serves prediction UI

Section 6 · The prediction model & measured accuracy

6.3 · The Prediction Model

Each dimension (length, width, height, weight) is predicted by models learned on the **log** of each measurement, keeping estimates positive and handling right-skewed pack sizes.

- **ExtraTrees** — point estimate + a data-driven confidence band from tree spread
- **HistGradientBoosting** — averaged in to lower point-estimate error
- **Weighted k-NN** — surfaces the closest real historical packs

The final estimate blends the ML model with nearest-neighbour in a **65 / 35** ratio — the most reliable combination on held-out data.

FEATURES THE MODEL READS

- **Numeric** — strength, concentration, fill volume, content weight, dose / strip / unit / bulk counts
- **Categorical** — curated Category and container type
- **Text** — name, brand, pack & strength, hashed to buckets

EACH PREDICTION RETURNS

- L · W · H (mm), volume (cm³), gross weight (g)
- Confidence score + likely range per dimension
- Closest reference packs + interactive 3D / 2D views

6.4 · MEASURED ACCURACY (HELD-OUT PACKS)

Dimension	Typical error	R^2
Length	± ~23 mm	~0.50
Width	± ~13 mm	~0.53
Height	± ~10 mm	~0.55
Weight	± ~63 g	~0.50

Accuracy improves as more real measurements are fed back in — every added, retrained pack becomes a new training example.

7 Conclusion

The Pharmaceutical Pack Size Prediction project turns thousands of scattered historical measurements into a single, reusable model that estimates a pack's dimensions and weight **before any physical sample is needed**.

By replacing repeated manual measurement with an instant, data-driven estimate, the solution speeds up carton, courier and warehouse planning while reducing measurement errors and duplicated effort.

After evaluating a full-framework build and a cloud-native enterprise build, the lightweight Python web application with a locally trained ML model is the most suitable option for current needs — cost-effective, transparent and easy to maintain, with a clear path to more advanced architectures as volumes grow.

Key Outcomes Expected

- Faster planning & quoting without waiting for samples
- Reduced manual measurement and re-measurement
- Consistent, traceable and reusable pack data
- Transparent estimates backed by real reference packs
- A confidence score guiding when to still measure
- A foundation that improves as data grows

8 References

LIVE APPLICATION
https://pharma-pack-size-prediction-web-app.onrender.com

DEPLOYMENT
Render web service · render.yaml · Python 3.12

SOURCE PROJECT
PHARMA_PACK_SIZE_PREDICTION_WEB_APP

SOURCE REPOSITORY LINK
https://github.com/Sarvesh-c07/PHARMA_PACK_SIZE_PREDICTION_WEB_APP.git

Core Project Files

`app.py`	Web server, prediction UI, 3D / 2D visualisation and ML-blend logic
`ml_engine.py`	The trained ExtraTrees + gradient-boosting model
`prepare_data.py`	Builds the model-ready training CSV from the master workbook
`train_model.py`	Cross-validates accuracy and rebuilds the cached model
`data/3S_Master_ML_Ready.xlsx`	Curated historical master dataset